

Homework 4

Due Date: August 27, 2019 by 5:00pm

MAT 022A Linear Algebra

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Instructions: Write solutions neatly and show all work. No need to show all steps when row reducing. Box your solutions, and explain your answers if necessary. Upload a PDF file of your assignment to Gradescope. When uploading to Gradescope, make sure to select which pages correspond to which question.

1. (Hill 6.2.8) Show that P_n is a subspace of P .

2. (Strang 3.1.17)

- a. Show that the set of *invertible* matrices in $\mathbf{M}$ is not a subspace.
 - b. Show that the set of *singular* matrices in $\mathbf{M}$ is not a subspace.
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3. (Hill 6.3.1) Which of the following sets of vectors span $\mathbb{R}^3$?

- a. $\{(1, -1, 2), (0, 1, 1)\}$
 - b. $\{(1, 2, -1), (6, 3, 0), (4, -1, 2), (2, -5, 4)\}$
 - c. $\{(2, 2, 3), (-1, -2, 1), (0, 1, 0)\}$
 - d. $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)\}$
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4. (Hill 6.3.10) Which of the following sets of vectors in $\mathbb{R}^3$ are linearly dependent? For those that are, express one vector as a linear combination of the rest.

- a. $\{(1, 2, -1), (3, 2, 5)\}$
 - b. $\{(4, 2, 1), (2, 6, -5), (1, -2, 3)\}$
 - c. $\{(1, 1, 0), (0, 2, 3), (1, 2, 3), (3, 6, 6)\}$
 - d. $\{(1, 2, 3), (1, 1, 1), (1, 0, 1)\}$
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5. (Hill 6.3.T.4) Suppose that $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set of vectors in a vector space V . Show that $T = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ is also linearly independent, where $\mathbf{w}_1 = \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3$, $\mathbf{w}_2 = \mathbf{v}_2 + \mathbf{v}_3$, and $\mathbf{w}_3 = \mathbf{v}_3$.

6. (Hill 6.6.1) Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$, where

$$\begin{aligned}\mathbf{v}_1 &= (1, 1, 2, 1), & \mathbf{v}_2 &= (1, 0, -3, 1), \\ \mathbf{v}_3 &= (0, 1, 1, 2), & \mathbf{v}_4 &= (0, 0, 1, 1),\end{aligned}$$

and $\mathbf{v}_5 = (1, 0, 0, 1)$. Find a basis for the subspace $V = \text{span } S$ of $\mathbb{R}^4$.

7. (Strang 3.4.26) Find a basis (and the dimension) for each of these subspaces of 3 by 3 matrices:

- a. All diagonal matrices.
 - b. All symmetric matrices ($A^T = A$).
 - c. All skew-symmetric matrices ($A^T = -A$).
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8. (Hill 6.5.12) Find a basis for the null space of the given matrix A .

$$A = \begin{bmatrix} 1 & -1 & 2 & 1 & 0 \\ 2 & 0 & 1 & -1 & 3 \\ 5 & -1 & 3 & 0 & 3 \\ 4 & -2 & 5 & 1 & 3 \\ 1 & 3 & -4 & -5 & 6 \end{bmatrix}$$

9. (Hill 6.6.10) Compute a basis for the row space of A , the column space of A , the row space of A^T , and the column space of A^T . Write a short description giving the relationships among these bases.

$$A = \begin{bmatrix} 2 & -3 & -7 & 11 \\ 3 & -1 & -7 & 13 \\ 1 & 2 & 0 & 2 \end{bmatrix}$$